

Threat Analysis and Risk Assessment Report

This report summarizes the identified attack paths, threat scenarios, impact ratings, and risk treatment decisions for the selected system.

1. Item Definition

Target Asset Door	Attack Vector Mode remote	Boundary External Vehicle Boundary
-----------------------------	-------------------------------------	--

2. Asset Identification

Asset: Door
Relevant Threat Count: 12
The target asset was selected from the TM7 item model and the threat scenarios were collected from the asset-to-threat mapping and the filtered attack graph result.

3. Impact Rating

Safety Major	Financial Major	Operational Moderate
Privacy Negligible	Maximum Impact Major	Impact Score 3

4. Damage Scenario Identification

Threat ID	Threat Name	Tactic	Damage Scenario
Threat_153	Control Doors	Affect Vehicle Function	This technique can directly cause damage to Door via remote access by enabling Control Doors, resulting in major impact on the target asset.
Threat_165	Vehicle Theft	Impact	This technique can directly cause damage to Door via remote access by enabling Vehicle Theft, resulting in major impact on the target asset.

5. Threat Scenario Identification

Threat ID	Threat Name	Tactic	Threat Scenario
Threat_24	Malicious SMS	Initial Access	This technique can be used as a threat scenario against Door via remote access and can contribute to a subsequent damage scenario that leads to major impact.
Threat_40	External Remote Services	Initial Access	This technique can be used as a threat scenario against Door via remote access and can contribute to a subsequent damage scenario that leads to major impact.
Threat_88	Network Sniffing	Credential Access, Discovery	This technique can be used as a threat scenario against Door via remote access and can contribute to a subsequent damage scenario that leads to major impact.
Threat_63	Exploit Co-Located Computing Device for Privilege Escalation	Privilege Escalation	This technique can be used as a threat scenario against Door via remote access and can contribute to a subsequent damage scenario that leads to major impact.
Threat_109	Exploit ECU for Lateral Movement	Lateral Movement	This technique can be used as a threat scenario against Door via remote access and can contribute to a subsequent damage scenario that leads to major impact.
Threat_116	Remote services in In-Vehicle Network	Lateral Movement	This technique can be used as a threat scenario against Door via remote access and can contribute to a subsequent damage scenario that leads to major impact.
Threat_108	Bridge Vehicle Networks	Lateral Movement	This technique can be used as a threat scenario against Door via remote access and can contribute to a subsequent damage scenario that leads to major impact.
Threat_115	Exploitation of Remote Services in In-Vehicle Network	Lateral Movement	This technique can be used as a threat scenario against Door via remote access and can contribute to a subsequent damage scenario that leads to major impact.
Threat_62	Abuse Elevation Control Mechanism	Privilege Escalation	This technique can be used as a threat scenario against Door via remote access and can contribute to a subsequent damage scenario that leads to major impact.
Threat_98	Network Service Scanning	Discovery	This technique can be used as a threat scenario against Door via remote access and can contribute to a subsequent damage scenario that leads to major impact.

6. Attack Path Analysis

6-A. Vehicle-Level Attack Paths (AI-Generated)

AI-generated attack path assessments for each UKC path. Each entry includes an attacker-perspective narrative, phase sequence, key assets, tactics, recommendations and required equipment.

DFD Attack Graph

Static image — interactive view available in browser.

No attack graph image generated.

No vehicle-level AI review available. Provide a Gemini API key to enable this section.

6-B. Functional-Level Threat Scenarios (AI-Generated)

Function-level threat scenarios derived from vehicle-level paths. Each scenario maps an attack path to a specific vehicle function, with feasibility ratings and SFOP impact assessment per ISO/SAE 21434.

No functional-level scenarios available. Provide a Gemini API key to enable this section.

7. Attack Feasibility Rating

Mode: remote

Derived CAL: CAL4

Impact Basis: Major

Threat ID	Threat Name	Feasibility	Risk Level	CAL	Risk Treatment Decision
Threat_24	Malicious SMS	High	High	CAL4	Reduce and define cybersecurity goals
Threat_40	External Remote Services	High	High	CAL4	Reduce and define cybersecurity goals
Threat_88	Network Sniffing	High	High	CAL4	Reduce and define cybersecurity goals
Threat_63	Exploit Co-Located Computing Device for Privilege Escalation	High	High	CAL4	Reduce and define cybersecurity goals
Threat_109	Exploit ECU for Lateral Movement	High	High	CAL4	Reduce and define cybersecurity goals
Threat_116	Remote services in In-Vehicle Network	High	High	CAL4	Reduce and define cybersecurity goals
Threat_108	Bridge Vehicle Networks	High	High	CAL4	Reduce and define cybersecurity goals
Threat_115	Exploitation of Remote Services in In-Vehicle Network	High	High	CAL4	Reduce and define cybersecurity goals
Threat_62	Abuse Elevation Control Mechanism	Low	Medium	CAL4	Reduce if feasible, otherwise retain with explicit rationale
Threat_98	Network Service Scanning	High	High	CAL4	Reduce and define cybersecurity goals
Threat_153	Control Doors	High	High	CAL4	Reduce and define cybersecurity goals
Threat_165	Vehicle Theft	High	High	CAL4	Reduce and define cybersecurity goals

8. Risk Value Determination

Internal Rule: Risk level is derived from *maximum impact* and *attack feasibility*. The score is calculated as $\text{Impact}(0-4) + \text{Feasibility}(0-4)$, then mapped to Low, Medium, High, or Critical.

- 0-2: Low
- 3-4: Medium
- 5-6: High
- 7-8: Critical

9. Risk Treatment Decision

For each identified threat scenario, the recommended treatment is shown in the table above. High or Critical risks should be reduced and traced to cybersecurity goals. Medium risks may be reduced or retained with an explicit rationale. Low risks may be retained with documented assumptions and monitoring.